# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** outside the circle | Compare 9 cm with the radius 5 cm. |
| 2 | **(c)** 4 | Midpoint of AB. |
| 3 | **(b)** 5 | Diagonals of a rectangle are equal and bisect each other. |
| 4 | **(d)** 60 | Triangle OAB is isosceles with OA = OB. |
| 5 | **(a)** 4 | Subtract the radii. |
| 6 | **(d)** equal and perpendicular bisectors of each other | Property |
| 7 | **(a)** 2 | Two pairs of equal opposite sides. |
| 8 | **(d)** 30 | $2(l + b)$. |
| 9 | **(b)** outside the circle | Compare 8 cm with the radius 5 cm. |
| 10 | **(c)** 7 | Midpoint of AB. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Compare 4 cm with the radius 6 cm (1); P lies inside the circle (1). | 1+1 |
| 12 | Diagonals are equal and bisect each other (1); OA = OB = 6 cm; BD = 12 cm (1). | 1+1 |
| 13 | Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 3 cm from A (1). | 1+1 |
| 14 | Compare 7 cm with the radius 7 cm (1); P lies on the circle (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Draw AB = 6 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3 cm from A (1). | 1+1+1 |
| 16 | Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1). | 1+1+1 |
| 17 | Diameter $= 8$ cm (1); Longest chord $= 8$ cm (1); No: 9 cm exceeds the diameter (1). | 1+1+1 |
| | **OR:** Draw AB = 7 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 3.5 cm from A (1). | 1+1+1 |
| 18 | Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1). | 1+1+1 |
| | **OR:** Diameter $= 12$ cm (1); Longest chord $= 12$ cm (1); No: 13 cm exceeds the diameter (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Draw AB = 8 cm (1); Draw perpendiculars at A and B and mark 6 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1). | 1+1+1+1+1 |
| | **OR:** P: inside (1); Q: on (1); R: outside (1); Inside or on the circle: 2 (2). | 1+1+1+2 |
| 20 | Draw AB = 12 cm (1); Draw perpendiculars at A and B and mark 5 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1). | 1+1+1+1+1 |
| | **OR:** P: on (1); Q: outside (1); R: inside (1); Inside or on the circle: 2 (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 8 | Diagonals of a rectangle are equal and bisect each other. |
| 2 | **(b)** 50 | Triangle OAB is isosceles with OA = OB. |
| 3 | **(d)** 4 | Subtract the radii. |
| 4 | **(b)** are equal but not perpendicular | Property |
| 5 | **(d)** 6 | It equals the side. |
| 6 | **(c)** 44 | $2(l + b)$. |
| 7 | **(c)** inside the circle | Compare 4 cm with the radius 6 cm. |
| 8 | **(b)** 9 | Midpoint of AB. |
| 9 | **(c)** 10 | Diagonals of a rectangle are equal and bisect each other. |
| 10 | **(b)** 40 | Triangle OAB is isosceles with OA = OB. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Diagonals are equal and bisect each other (1); OA = OB = 7 cm; BD = 14 cm (1). | 1+1 |
| 12 | Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 5 cm from A (1). | 1+1 |
| 13 | Compare 9 cm with the radius 5 cm (1); P lies outside the circle (1). | 1+1 |
| 14 | Diagonals are equal and bisect each other (1); OA = OB = 9 cm; BD = 18 cm (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Draw AB = 8 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 4 cm from A (1). | 1+1+1 |
| 16 | Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1). | 1+1+1 |
| 17 | Diameter $= 16$ cm (1); Longest chord $= 16$ cm (1); No: 17 cm exceeds the diameter (1). | 1+1+1 |
| | **OR:** Draw AB = 9 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 4.5 cm from A (1). | 1+1+1 |
| 18 | Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1). | 1+1+1 |
| | **OR:** Diameter $= 20$ cm (1); Longest chord $= 20$ cm (1); No: 21 cm exceeds the diameter (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Draw AB = 6 cm (1); Draw perpendiculars at A and B and mark 4 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1). | 1+1+1+1+1 |
| | **OR:** P: on (1); Q: inside (1); R: outside (1); Inside or on the circle: 2 (2). | 1+1+1+2 |
| 20 | Draw AB = 15 cm (1); Draw perpendiculars at A and B and mark 8 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1). | 1+1+1+1+1 |
| | **OR:** P: outside (1); Q: on (1); R: inside (1); Inside or on the circle: 2 (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Playing with Constructions** | Class 6 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 7 | Subtract the radii. |
| 2 | **(a)** perpendicular bisector of AB | Property |
| 3 | **(d)** 8 | It equals the side. |
| 4 | **(a)** outside the circle | Compare 5 cm with the radius 4 cm. |
| 5 | **(c)** 12 | Midpoint of AB. |
| 6 | **(b)** 13 | Diagonals of a rectangle are equal and bisect each other. |
| 7 | **(b)** 30 | Triangle OAB is isosceles with OA = OB. |
| 8 | **(d)** 4 | Subtract the radii. |
| 9 | **(b)** 10 | It equals the side. |
| 10 | **(b)** inside the circle | Compare 2 cm with the radius 7 cm. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 8 cm from A (1). | 1+1 |
| 12 | Compare 3 cm with the radius 8 cm (1); P lies inside the circle (1). | 1+1 |
| 13 | Diagonals are equal and bisect each other (1); OA = OB = 11 cm; BD = 22 cm (1). | 1+1 |
| 14 | Draw the perpendicular bisector with compass arcs of the same radius above 'half AB' (1); The point is 10 cm from A (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Draw AB = 10 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 5 cm from A (1). | 1+1+1 |
| 16 | Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1). | 1+1+1 |
| 17 | Diameter $= 24$ cm (1); Longest chord $= 24$ cm (1); No: 25 cm exceeds the diameter (1). | 1+1+1 |
| | **OR:** Draw AB = 12 cm (1); With A and B as centres and the same radius more than half of AB, draw arcs on both sides (1); Join the intersection points; the line meets AB at its midpoint, which is at 6 cm from A (1). | 1+1+1 |
| 18 | Diagonals of a square are perpendicular (1); $\angle AOB = 90^\circ$ (1); Triangle AOB is isosceles right-angled so $\angle OAB = 45^\circ$ (1). | 1+1+1 |
| | **OR:** Diameter $= 30$ cm (1); Longest chord $= 30$ cm (1); No: 31 cm exceeds the diameter (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Draw AB = 9 cm (1); Draw perpendiculars at A and B and mark 4 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1). | 1+1+1+1+1 |
| | **OR:** P: outside (1); Q: on (1); R: inside (1); Inside or on the circle: 2 (2). | 1+1+1+2 |
| 20 | Draw AB = 7 cm (1); Draw perpendiculars at A and B and mark 5 cm on each (1); Join the free ends (1); Draw both diagonals and measure: AC = BD (1); Diagonals bisect each other so OA = OC = AC/2 (1). | 1+1+1+1+1 |
| | **OR:** P: inside (1); Q: on (1); R: outside (1); Inside or on the circle: 2 (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.